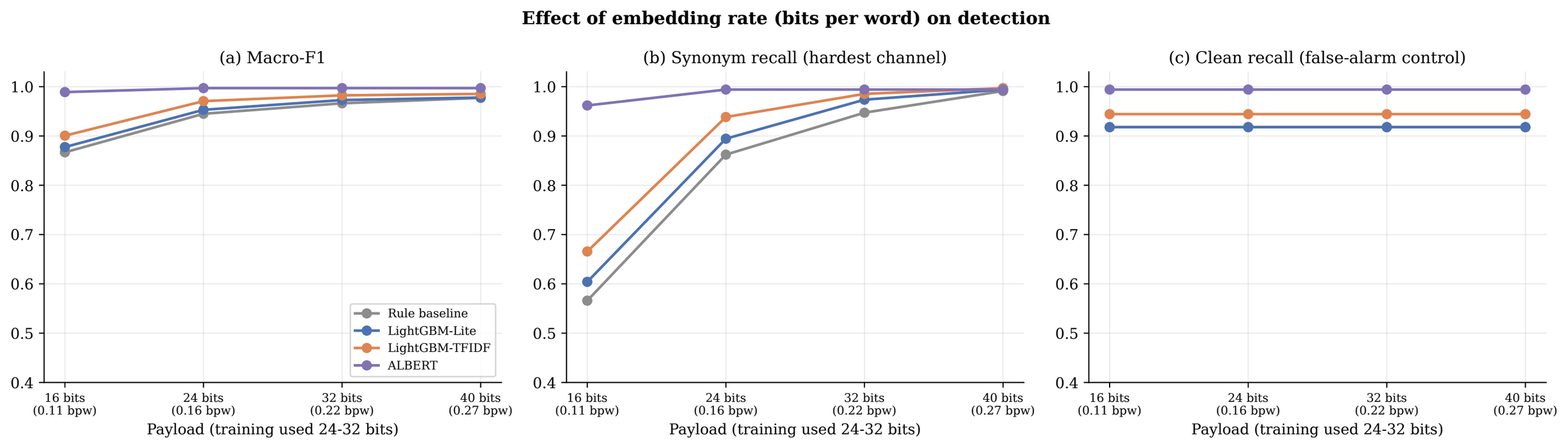
Effect of embedding rate (bits per word) on detection
(a) Macro-F1
(b) Synonym recall (hardest channel)
(c) Clean recall (false-alarm control)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Rule baseline
LightGBM-Lite
LightGBM-TFIDF
ALBERT
16 bits
(0.11 bpw)
24 bits
(0.16 bpw)
32 bits
(0.22 bpw)
40 bits
(0.27 bpw)
Payload (training used 24-32 bits)